Management of normal labor

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Objectives

⑩ **At the End of this Lecture Each Student should be able to:-**

1-Discuss factors affecting the decision of admission.

2-obtain complete history of the mother before admission.

3-Performe examination of the abdomen.

4-Performe examination of the vagina

5- Obtain laboratory testes on admission.

6-Apply management of the first stage.

7-Prepare the delivery room.

8-Understand duties of the scrub nurse.

9- Understand duties of the circulating nurse

10- Apply management of second stage of labor.

11- Apply management of third stage of labor

12- Apply management of fourth stage of labor

Admission to labor unit

Factors Affecting the Decision of Admission

1- Frequency of contractions (3/10 admit)

2- Prior vaginal deliveries (admit)

3- Cervical dilatation (>4cm admit)

4- Condition of membranes (rupture admit)

Obtaining the Admission History and Performing Examinations

Identifying Information

- Name
- Age
- Duration of marriage
- Educational level
- Last meal since

History of Past Pregnancy

- Mode of delivery
- Size of infant
- Birth intervals
- Duration of pregnancy
- Congenital defects
- Any abnormality affecting that pregnancy.

Medical and Family History

- Major medical & or surgical problems
- Drug allergies
- **History of Current Pregnancy:-**
- Prenatal care
- Laboratory tests
- Special tests as amniocentesis, NST.
- Details of any complications or problems and their treatment

OB .History

- LMP
- EDD or EDC
- gestational age
- Parity, gravida.
- Last labor since
- Mode of previous delivery
- Any Medical or OB. Complications has attacked the present pregnancy
- Fetal condition (Biophysical Profile)

Examination of the Abdomen

- Inspect the contour of the abdomen.
- Palpate the abdomen gently with warm hands, for the height of the fundus, lie, presentation, position, attitude and station then record them.
- Observe and record the frequency, duration, and intensity of uterine contractions.
- Auscultate and record the fetal heart sound for one minute using Pinard's or Sonicade.

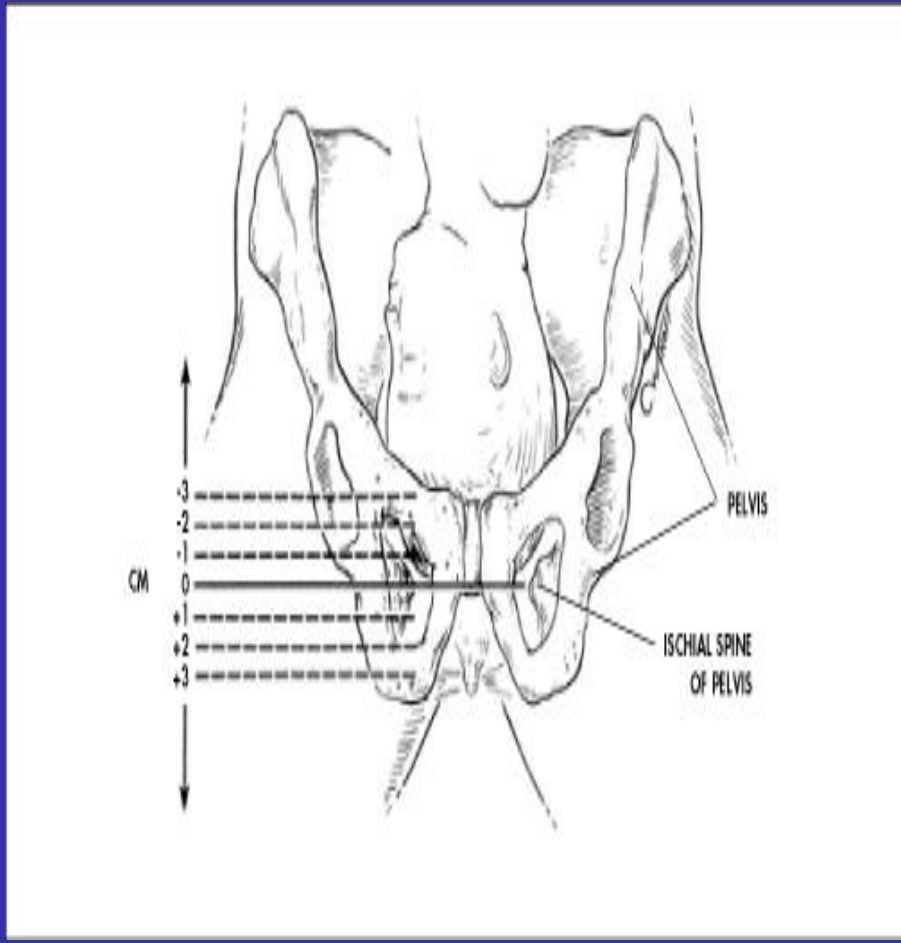
Examination of the Vagina

1-Assess the progress of labor.

2-Determine the extent of effacement and the degree of dilatation of the cervix.

3-Assess the condition of the membranes.

4-Determine position, presentation and degree of descent of head in pelvis in relation to ischial spine (station).



Cont. Examination of the Vagina

5-Observe the presence of umbilical cord beside the head.

6-Check moulding, presence of caput succedaneum.

7-Evaluate pelvic capacity

Obtaining Laboratory Testes

- ABO/ Rh.
- CBC
- Urine analysis: measure for glucose and proteinuria.

General Management of the first stage of labor

1. Asepsis:

The danger of infection has been minimized by the use of antibiotics, it is very important to take all antiseptic measures

2. Examination of the patient:

- General, abdominal and vaginal examinations are carried out
- Urine sample is taken for sugar and albumin

3. The urinary bladder:

Kept empty by frequent micturation or passing a catheter at strict antiseptic measures

4. The rectum:

Evacuated by low enema (warm water), to avoid inertia and soiling of the perineum by feces during delivery (new trends)----- must be avoided.

5. Patient position:

- The patient is allowed to walk if the head is engaged.
- Rest in bed is recommended when membranes are ruptured or the head not engaged

6-Diet: Nothing is given by mouth except sugary fluid to avoid vomiting. I.V glucose is given in prolonged labor

7. Sedative and analgesics: To avoid exhaustion from pains e.g. pethedine 50 mg IM

8. Observations:

a. Maternal observations

- Pulse, temperature, blood pressure every hour
- Uterine contractions: Frequency and duration from 1/2 - 1 hour
- Time of rupture of membranes
- Degrees of cervical dilatation

b. Fetal observations:

- Fetal heart sound: every 30 minutes
- Degrees of head descend

Nursing Care during First Stage Labor

- 1- Assessment
- 2-Emotional/Supportive

1- Assessment

1-Monitor and follow up of the following:-

	Latent phase	Active acceleration Phase	Active deceleration Phase
Vital signs	4hour	2 hour	1 hour
Uterine contractions	1 hour	30min.	15min.
Behavior pattern	1 hour	30min.	15min.
Fetal heart rate	1 hour	30min.	15min.
Vaginal examination	1 hour	30min.	15min.

2-Nutrition: IV glucose or ringer lactates to prevent dehydration

3-Nonpharmacological interventions

- Assist client to breathe through each contraction
- Assist with relaxation techniques between contractions to avoid maternal exhaustion.
- Provide encouragement to client during labor.

- If possible, have a family member or friend stay with client during labor.
- Provide sips of water or juice during early labor.
- If client is unable to tolerate fluids orally, provide mouth care with water swabs to coat the lips.

Physical sensations during latent phase

- Thirst, anorexia, nausea, vomiting, diarrhea

- Uterine contractions usually become more frequent and intense as latent labor progresses; felt in the abdomen and lower back

- Fetal descent may cause pressure on bladder, creating urge to urinate

- Fatigue if latent labor is prolonged.

2- Emotional Support

Latent Phase

provide current status and process of labor

Demonstrate breathing and relaxation

techniques and work with support person

2-Emotional Support

Active Phase

-feedback about performance; work with support person

-Reduce distractions (quiet)

-Review current status and process of labor

-Position changes

2-Emotional Support

Transition Phase

-Client may lose control; provide continuous support

-Reduce distractions (quiet)

-Help to relax between contractions

-Inform client about short duration of transition phase

-Keep informed of progress

Nursing Care During Second Stage Labor

<u>Assessment</u> -- Assessment is continuous during second stage labor

Assessment Item	Frequency
Vital signs (BP, P, R)	q 5-30 minutes
Uterine activity	Every contraction
Bearing down effort	Every contraction
Fetal heart rate	q 5 minutes
Signs of fetal descent	q 10 minutes
Behavior, mood, etc.	q 10-15 minutes

Preparation of the delivery room

1- The delivery room should be warm

2- Enough privacy should be provided

3- Delivery equipments (2 kokher, 2 arteries, 1 needle holder, 2 scissors, 1 tissue forceps,drugs as methergine,oxytocin,catgut)

4- Resuscitation equipment

5- Warm Incubator

6- Oxygen unit

7- Infant suction apparatus

8- Infant resuscitation bag and masks.

Preparation of the woman

1-Transfer to delivery room, should be between contractions and supported adequately (primipara when cervix is fully dilated; multipara when cervix is 8cm).

2- Posture of the woman (dorsal or lithotomy; left lateral)

3- The legs and thighs should be dressed with sterile leggings

4- Sterile towels put over the abdomen and under buttocks leaving only the vulva and perineum exposed.

Techniques of second stage

A. Delivery positions

Dorsal lithotomy: provides:

- Maximum perineal exposure

Dorsal lithotomy

Preparation of the attendance

1-The attendance should wear an head cap and mask,

2-Wash and scrub hands

3- Put on a sterile gown and gloves.

Duties of the scrub nurse

1-Arranges the instruments on the delivery table

2- Assists in draping the woman

3- Handles the instruments and catheter to the doctor as called for

4- Assists in perineal repair and prepares proper sutures on proper needles

5- Massages the uterus as needed

6- Assists in holding the newborn infant.

Duties of the circulating nurse

1-Assists the woman during her transfer from the labor bed and positions her properly on the delivery table

2- Opens packs

3- Prepares and administers routine medications as ordered and records its time

4- Prepares resuscitation and emergency equipments.

5- Count and records FHR

6-Receives the infant from the doctor and takes care of him

7- Records the time of birth and sex of the infant

8- Observes and records the time of placental expulsion

9- Observes the location and extent of episiotomy or tear

10- Records the amount of blood loss

11- Observes and records any unusual conditions

12- Prepares and places the identification measure of the infant

13- Assists in giving anesthesia and emergency drugs if needed

Care during delivery

1-Encourage mother to take deep breath, holds it and bear down during contractions and to relax between contractions.

2-Support perineal to prevent laceration

3- Once head is delivered, check for presence of cord around neck

4-If cord is wrapped around neck, gently slip a finger under cord and pull it overhead

After delivery of the head

1-Wipe face of secretion

2- Suction nose and mouth

Management of the Third stage of labor

- Apply two forceps on umbilical cord and cut in between
- Carefully inspect vagina, perineum and labia for tears or lacerations
- Inspect vulva for bleeding, edema and hematoma
- Make sure that the uterus is well contracted.
- Avoid vigorous uterine massage

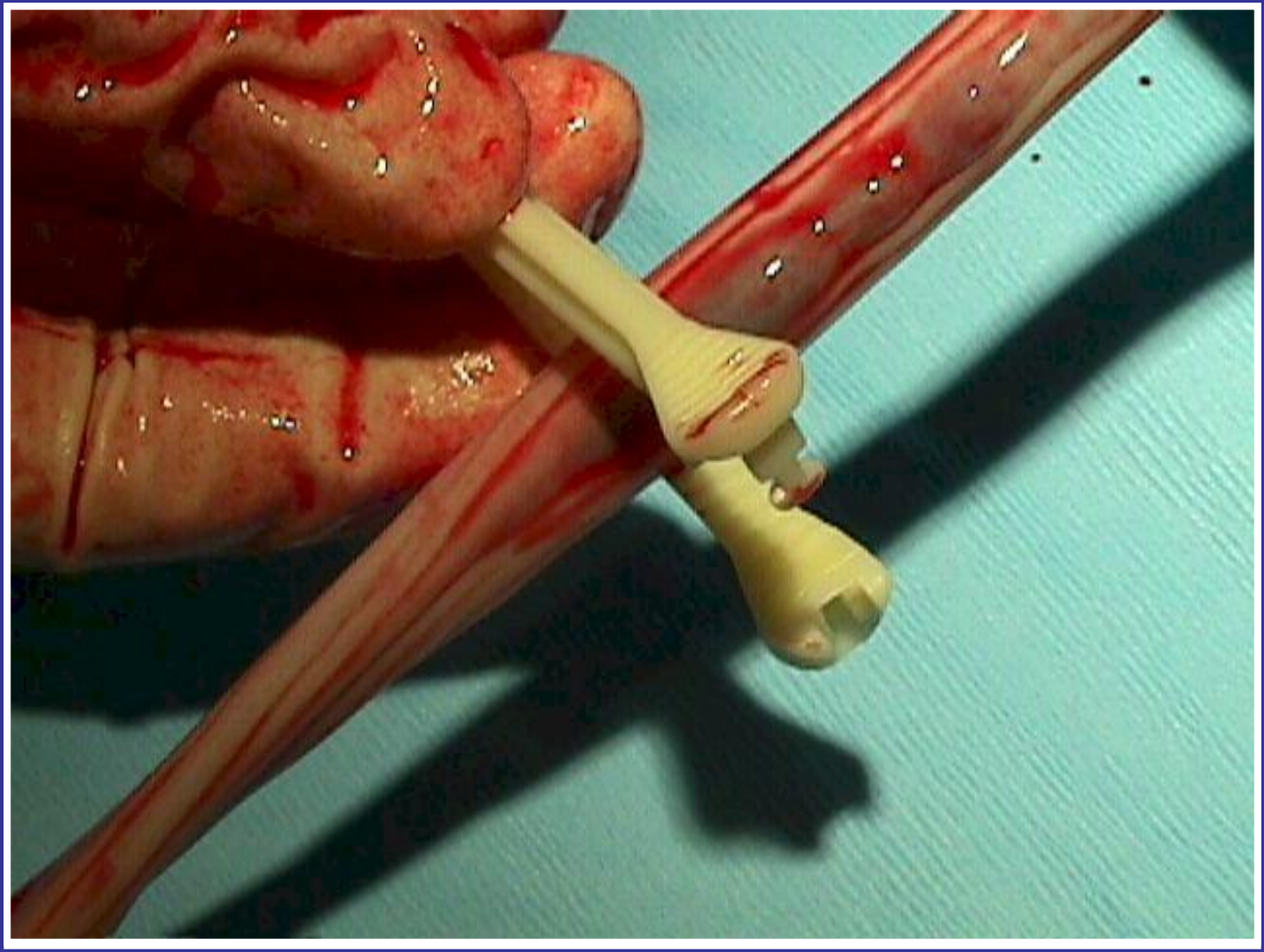
- ⑩ The placenta is expelled by controlled cord traction and use of oxytocic drugs
- ⑩ Receive placenta in the hands and turns through a circle
- ⑩ Examine the placenta and membranes for general appearance, completeness and weight
- ⑩ Clean and dry the vulva, buttocks and thighs then apply a sterile pad.

The umbilical cord

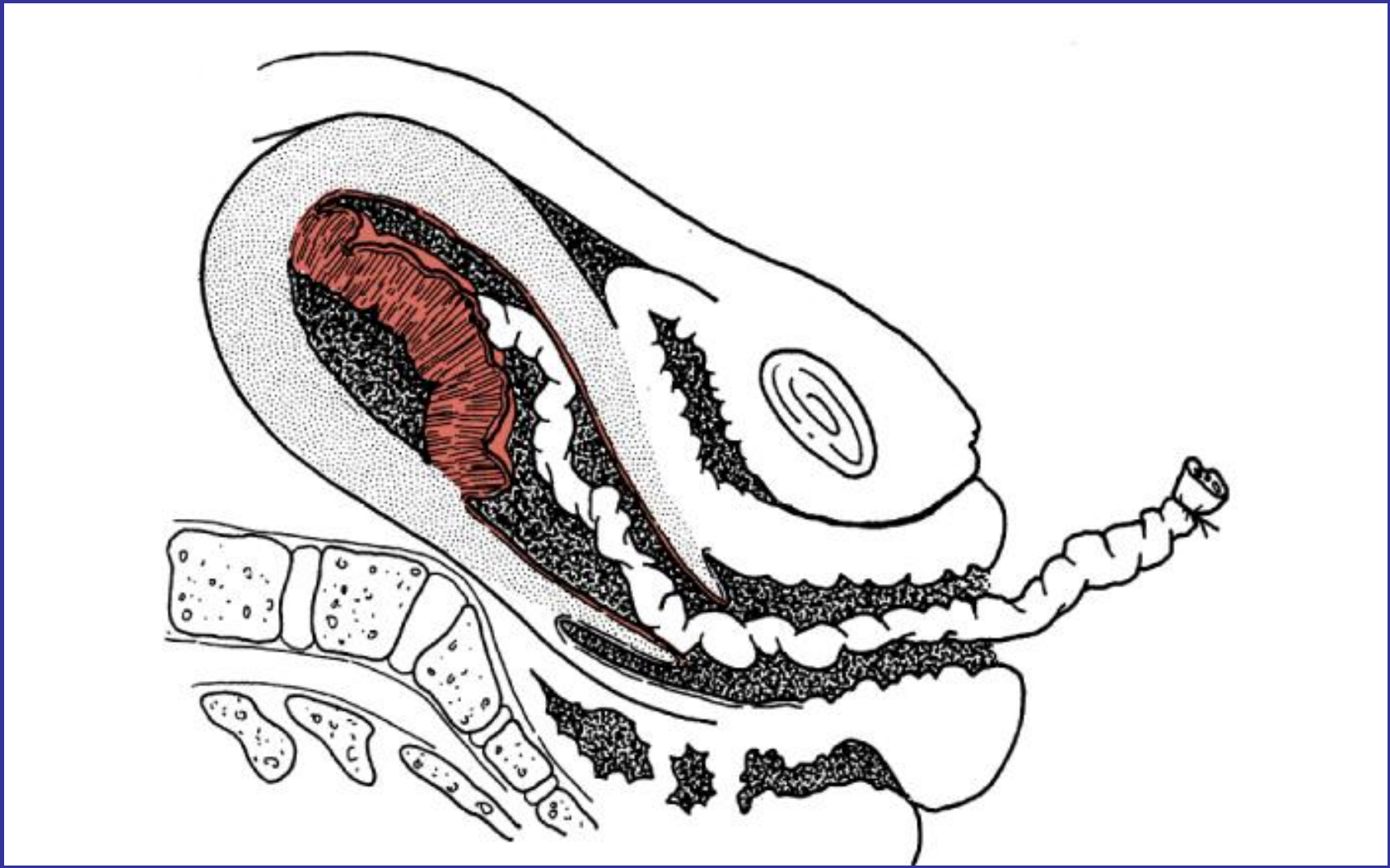
The cord is massaged towards the fetus several times this brings 60:90 cc blood from placenta.

When pulsation ceases the cord is cut off between 2 clamps

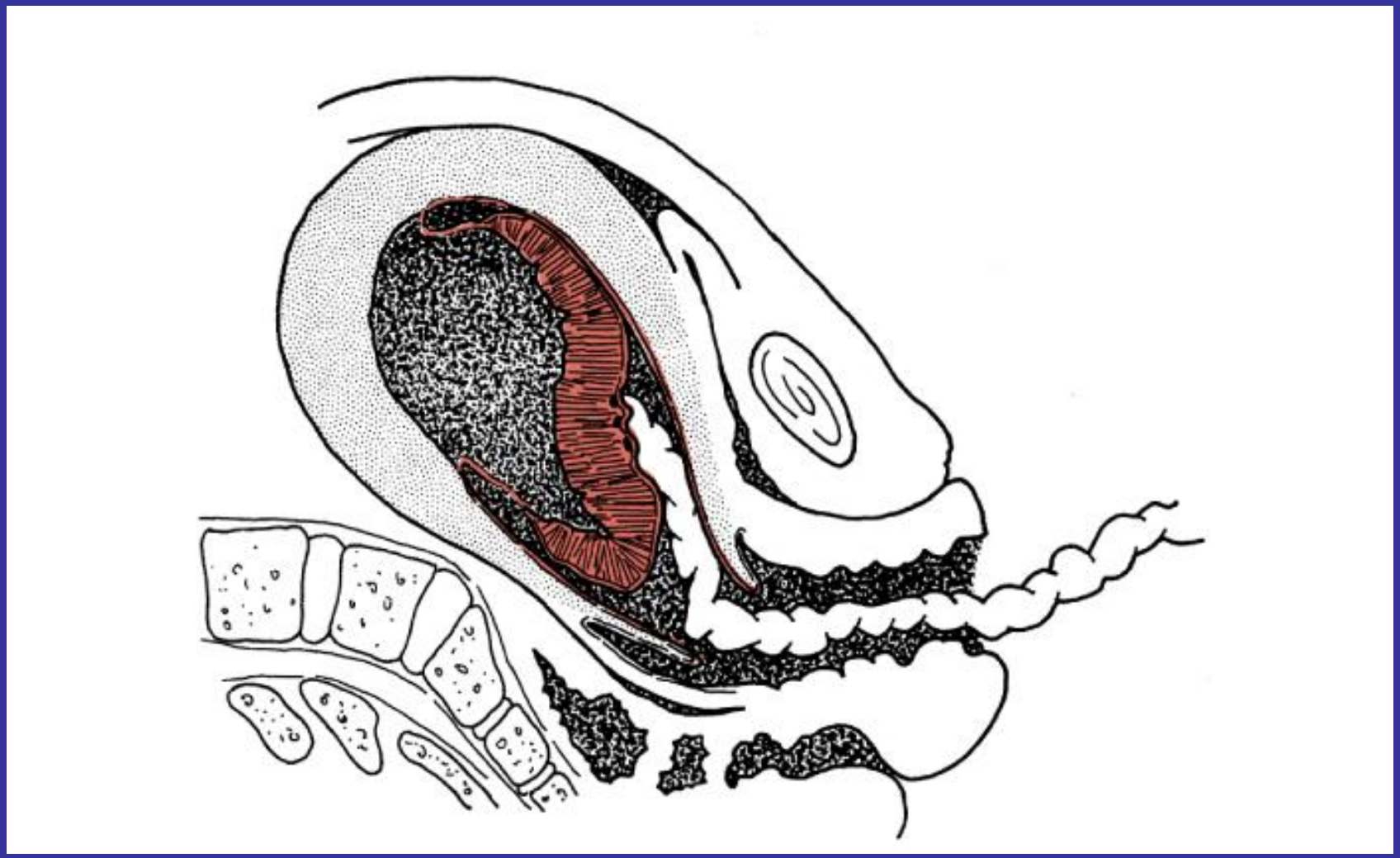
The umbilical cord



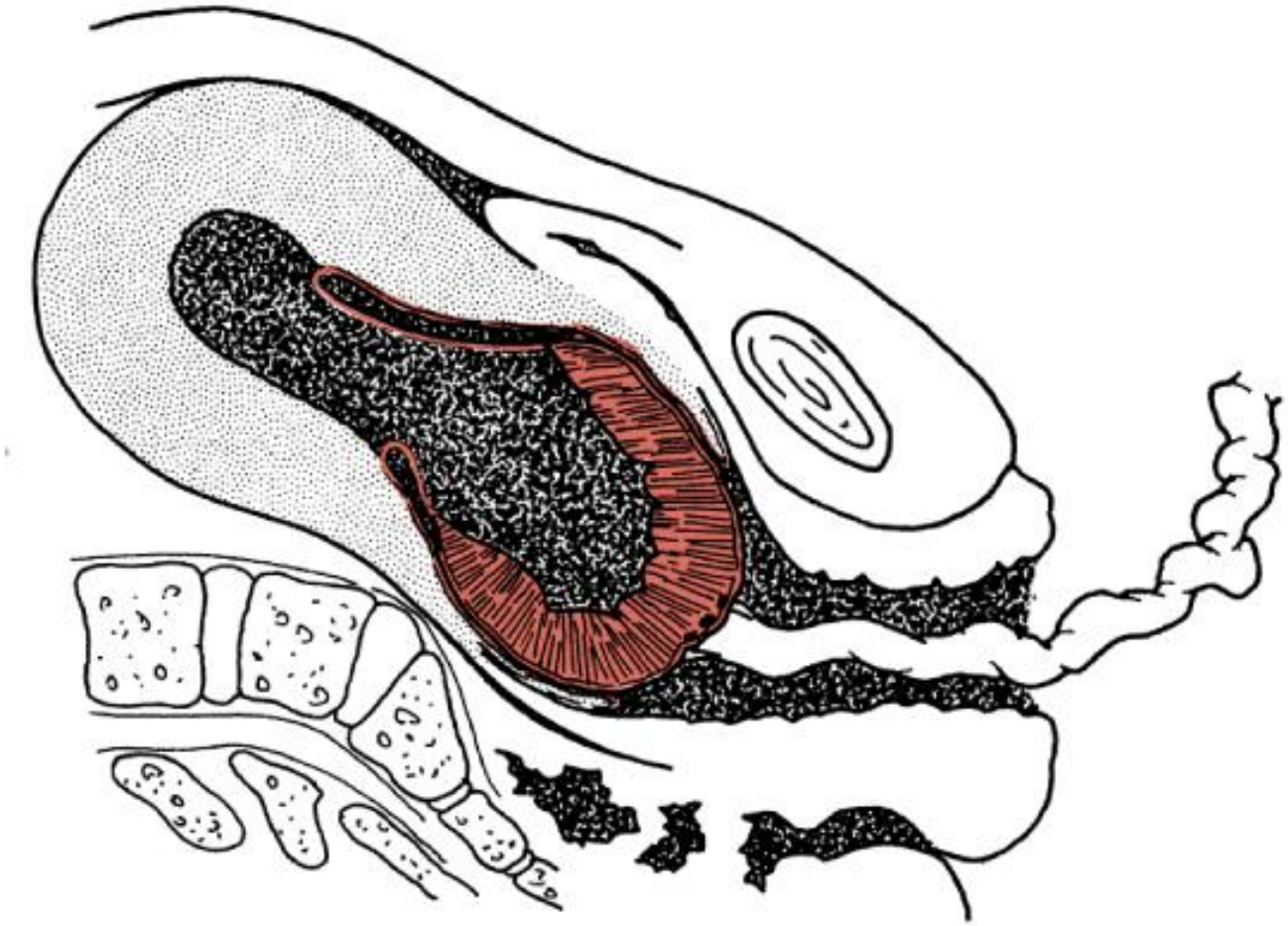
Placental Separation (Beginning)



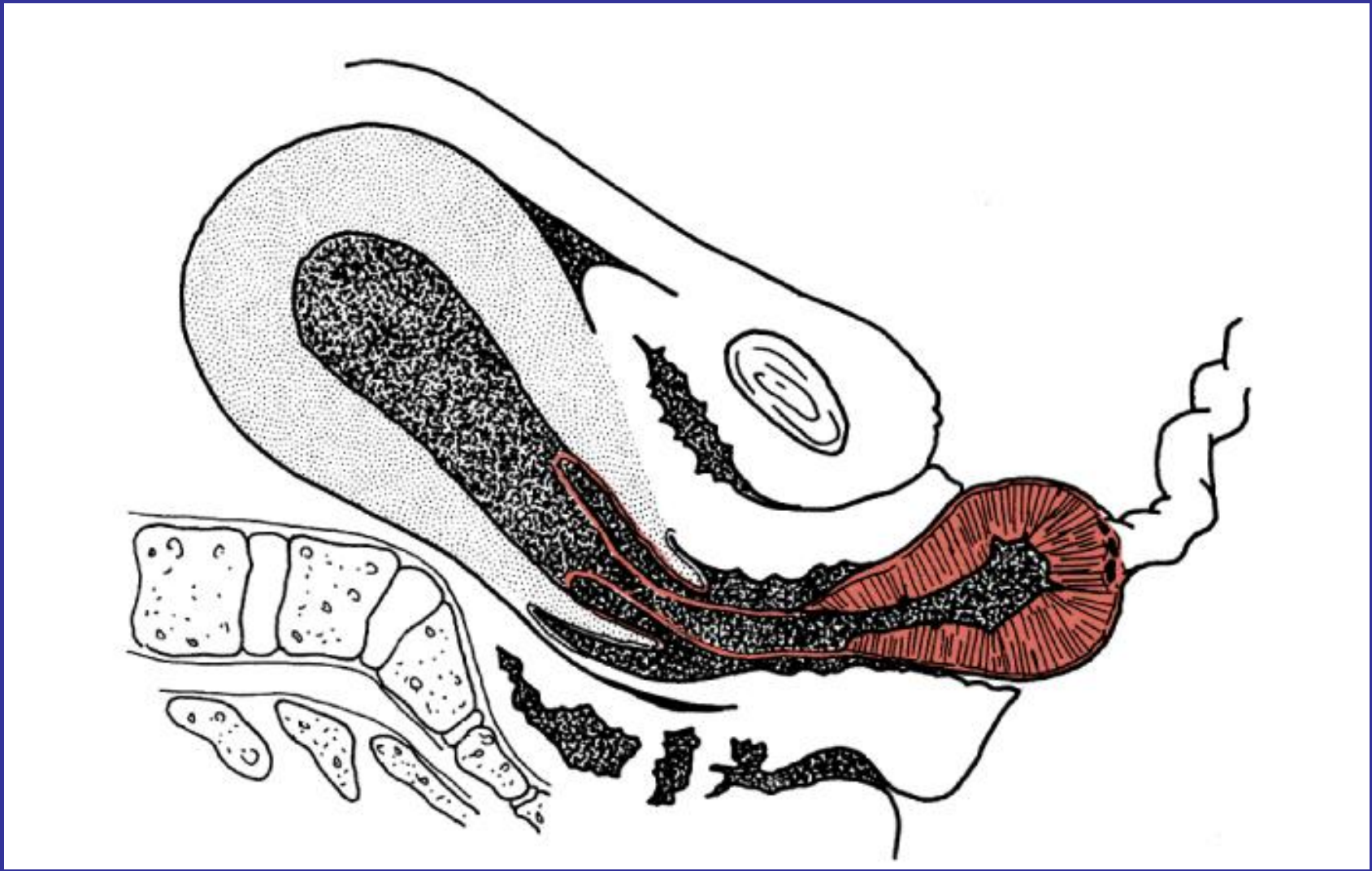
Placental Separation (Complete)



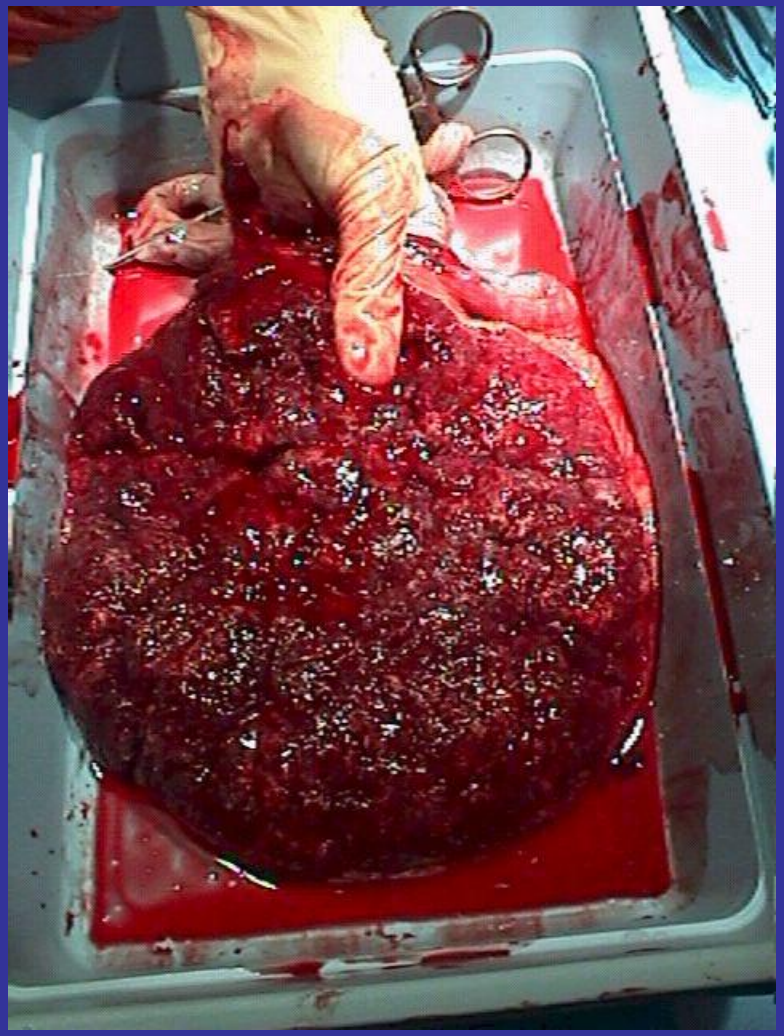
Placental Expulsion (Early)



Placental Expulsion



Inspect the Placenta



- Make sure it is complete
- Look for missing pieces
- Look for malformations
- Look for areas of adherent blood clot

Nursing role in immediate newborn care

The main nursing managements are:

Clear airway.

Maintain of body temperature.

Cutting of cord

Eye care

Identification.

Apgar Score

Immediate need for resuscitation

	0	1	2
Heart Rate	Absent	< 100 bpm	> 100 bpm
Respiratory Effort	Absent	Slow, weak cry	Good cry
Muscle Tone	Flaccid	Some flexion of extremities	Well-flexed
Reflex Irritability	No response	Grimace	Cry
Color	Blue, pale	Body pink, extremities blue	Completely pink

Implications of Apgar Score

- 0 – 3 Severe distress.. Requires emergent resuscitation measures.
- 4 – 6 Moderate difficulty.. Requires supplemental oxygen and appropriate resuscitation measures.
- 7 – 10 No difficulty.. Is considered normal.

Management of the Fourth Stage of Labor

-Check and record maternal vital signs every 15 minutes

-Check the uterus to ensure that it is well contracted to prevent bleeding

-Observe and record the amount of bleeding

-Inspect the perineum for edema or hematoma

-Encourage the woman to pass urine

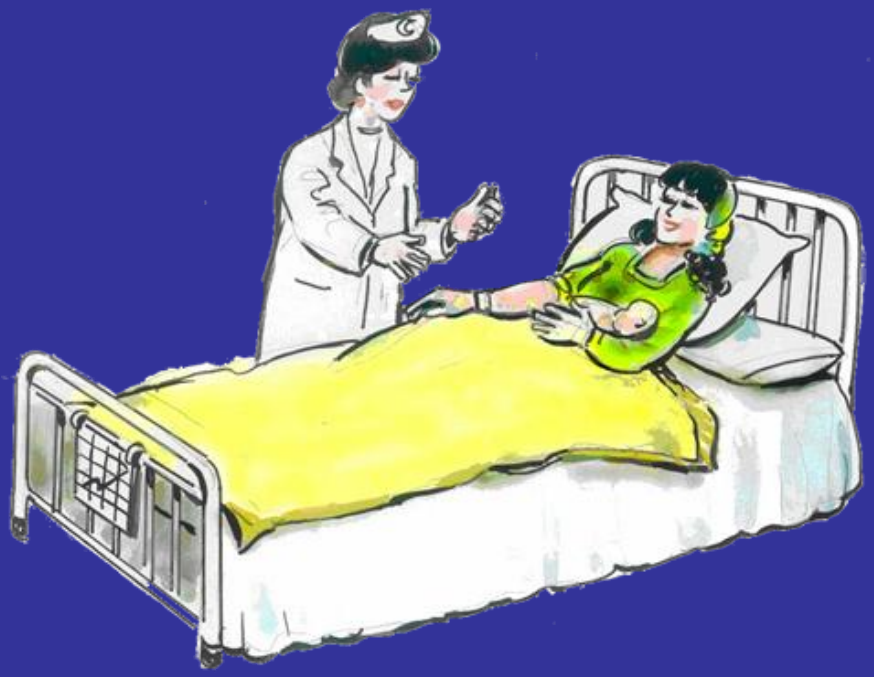
-Keep woman dry and clean

-Light snack and drink

-Encourage rest and sleep

END OF LECTURE

THANKS///////



Second stage of labor

It is the time of greatest physical strain for the mother and the fetus "Longest journey in the world" 6 cm from ischeal spines to the perineum.

Phases of the second stage:

Descend phase:

During descend the head rotates, the vagina becomes dilated, and the mother develops an urge to push

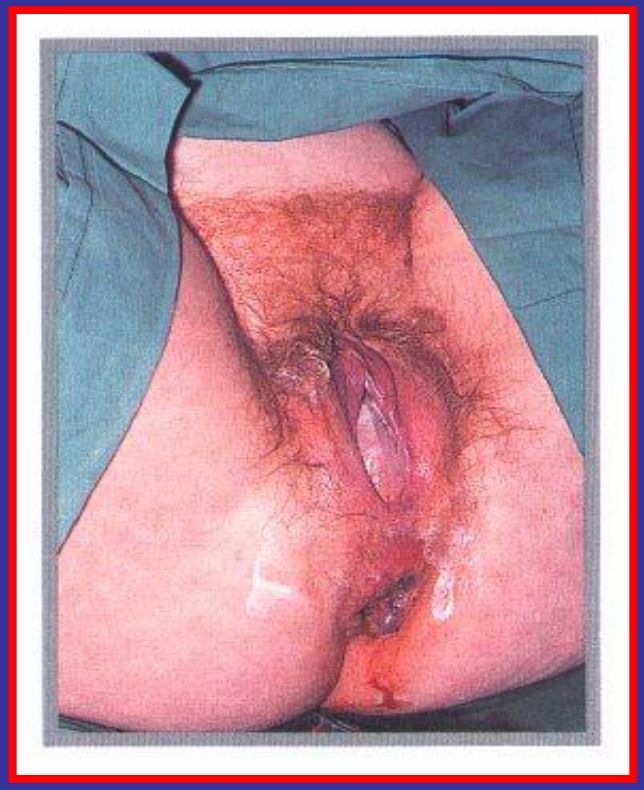
Outlet phase :

Head on the thin pelvic floor and becomes visible, **"crowning"** occurs

Delivery of the head

When the head appears at the vulva, the perineum is supported during contractions by a sterile pad to keep the head flexed till crowning.

After crowning the head is delivered by sliding the perineum over the face in-between contractions not at the summit of contractions.

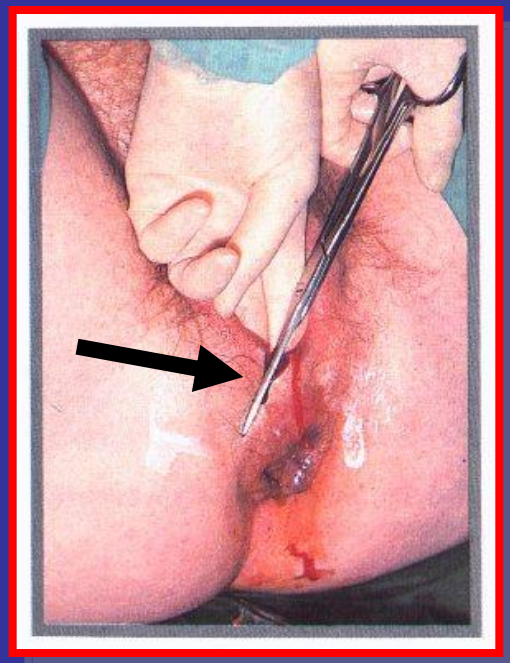


Delivery of the head

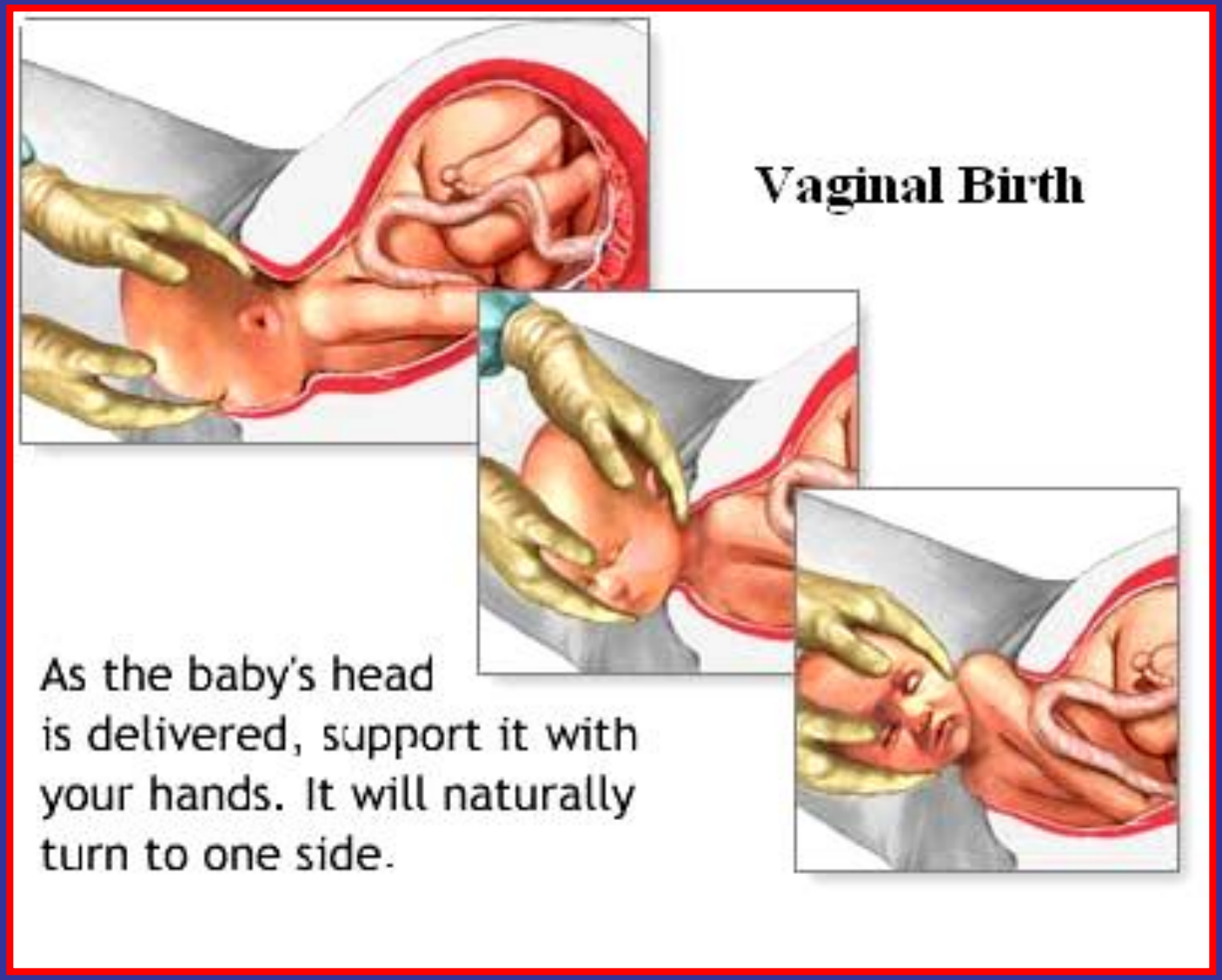
Episiotomy is performed when the perineum is threatened to rupture

After delivery of the head, the mouth and nose are cleaned with sterile piece of gauze to remove mucous

The neck is inspected if the cord is coiled around, try to slip it , if failed the cord is divided between 2 clamps



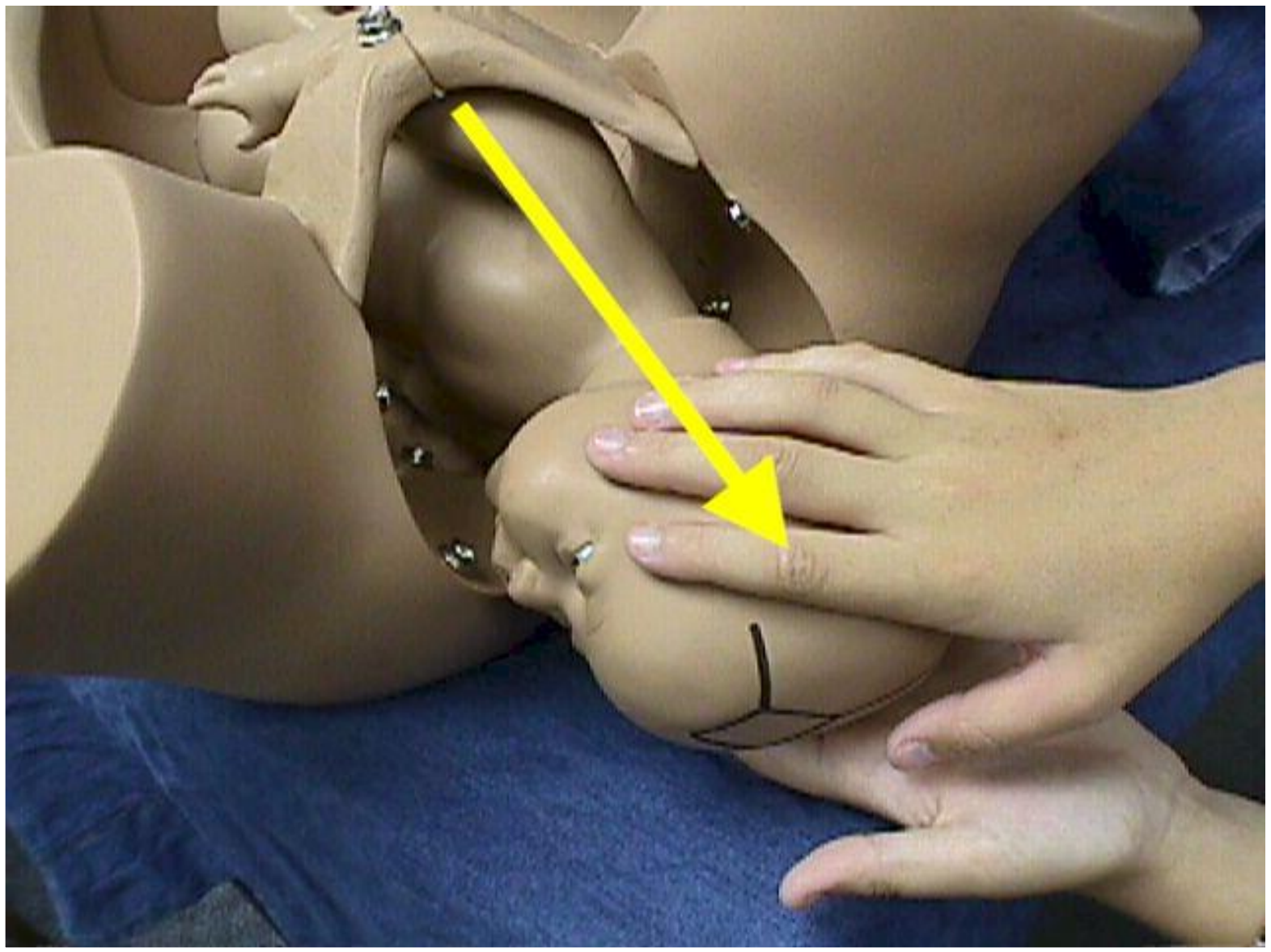
Second stage of labor



D. Delivery of the shoulders

- The head is grasped between 2 hands (After external rotation has been completed), traction is applied downwards and backwards till the anterior shoulder appears below the symphysis pubis.
- Then the head is elevated upwards to deliver the posterior shoulder.
- The trunk is delivered by gentile traction

D. Delivery of the shoulders



Crowing

Definition:

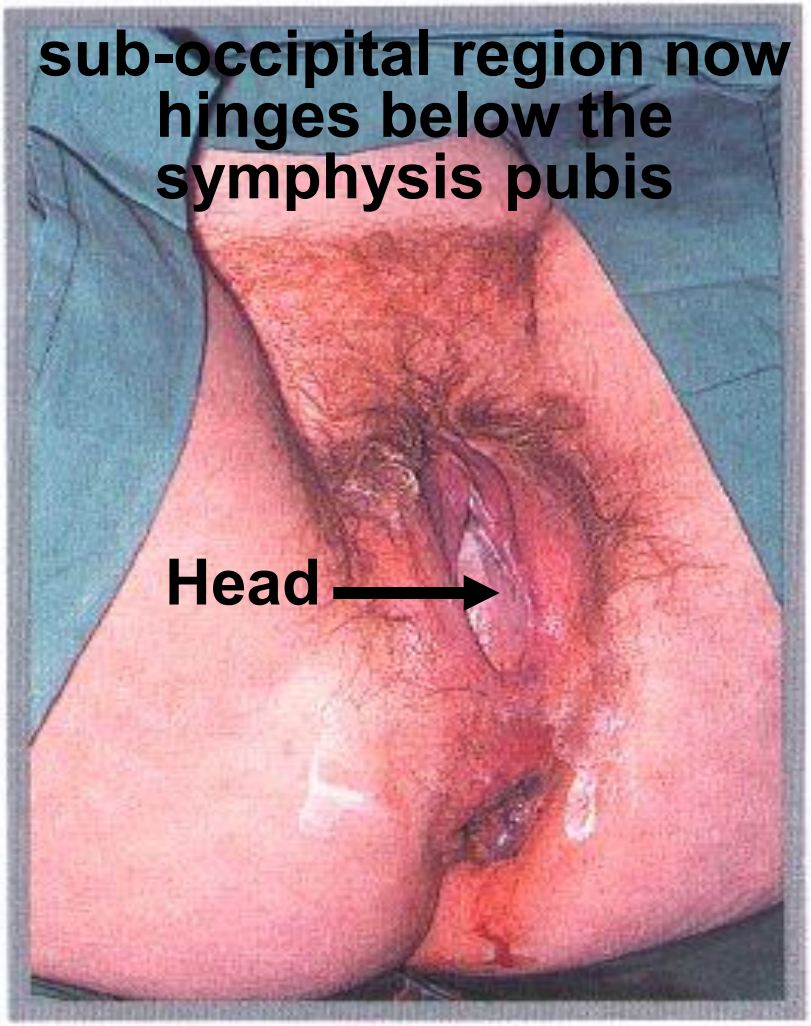
The bi-parital diameter of the fetal head had just passed outside the vulval ring and not recedes back.

So the sub-occipital region now hinges below the symphysis pubis

Value:

If the head is allowed to be delivered before crowning extension occurs and head passes by the occipito-frontal diameter (11.5 cm) which causes perineal tear, so the head must be supported till crowning to allow delivery by suboccipito-bregmatic diameter

Crowing


sub-occipital region now hinges below the symphysis pubis
Head

C. Management of the third stage

Signs of placental separation:

The hand is placed over the fundus to detect signs of placental separation and descend

1. The body of uterus becomes smaller, harder and more globular (the earliest sign)
2. The fundus rise up because the LUS now distended by the placenta
3. A supra-pubic bulge due to presence of placenta in LUS
4. Elongation of the cord outside the vulva
5. Sudden gush of blood from the vagina

C. Management of the third stage

- When the placenta appears at the vulva it is rolled between 2 fingers to make a rope of membranes, then the placenta and membranes are examined
- If fragment is missing the uterus is explored under general anesthesia
- The genital tract is explored if any lacerations are detected it should be sutured immediately or within 24 hours otherwise it is considered as septic or potentially septic wounds

C. Management of the third stage

Repair of episiotomy and any perineal lacerations immediately after placental delivery

Methods of placental delivery

a. The conservative methods:

- After delivery of the fetus, the left hand is placed over the fundus to detect signs of placental separation and descend.
- When these signs appear the uterus is massaged to contract and the patient is asked to bear down to deliver the placenta.
- This can be helped by gentle traction on the cord and 0.5 mg ergometrin given I.V or I.M given after placental delivery to diminish bleeding

Methods of placental delivery

b. The active method:

- With delivery of the fetus, ergometrin 0.5 mg is given IV or IM or Syntometrin vial I.M.
- The placenta is then delivered by Brandt-Andrews or Crede's methods (see retained placenta).
- The advantages of active methods are to avoid postpartum hemorrhage but ergometrin may cause contraction ring and retained placenta.